

NYC Taxi Mobility and Safety Trends (2024)

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Project Overview

This report presents an in-depth data storytelling analysis of New York City's yellow taxi system in 2024. Using multiple open datasets including taxi trip records, traffic volume logs, MTA subway ridership data, and monthly crash reports, we developed four themed dashboards to explore emerging patterns in urban mobility, pricing, demand dynamics, and safety. Our insights reveal how daily, hourly, and seasonal factors interact with pricing strategies and traffic behavior, ultimately shaping the urban transportation experience for riders and drivers alike.

Motivation: Why We Chose This Topic

New York City is one of the busiest and most complex urban transportation ecosystems in the world. Yellow taxis have long been a symbol of the city's movement and accessibility. However, with the rise of ride-sharing platforms, fluctuating traffic congestion, and evolving public transportation usage, it's crucial to understand how the traditional taxi system is adapting. We wanted to explore key questions such as: When and where are taxis most in demand? Do fare patterns align with rider behavior? How does seasonal travel impact safety? And which areas of the city are under the most traffic pressure?

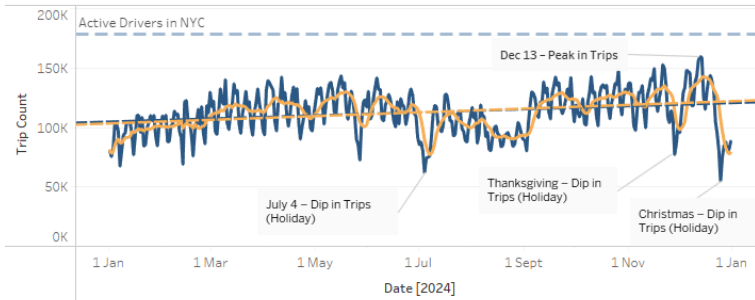
This project is motivated by a desire to use data to understand not only how people move across NYC, but also how systems like pricing and infrastructure respond to that movement. Through a visual storytelling approach, we aim to bring clarity to these questions for planners, policymakers, and curious citizens alike.

Dashboard 1: Citywide Mobility Overview

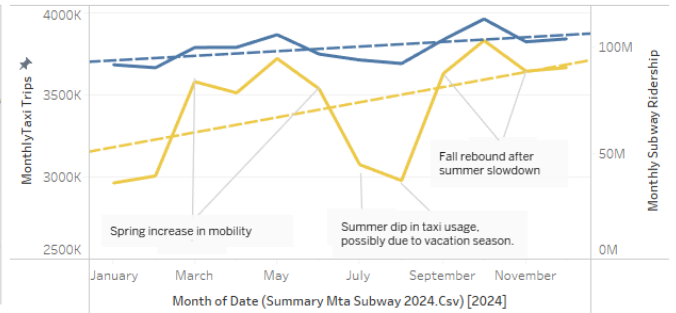
Citywide Mobility Overview

Understanding NYC's transportation pulse before diving into pricing, peaks, and safety patterns.

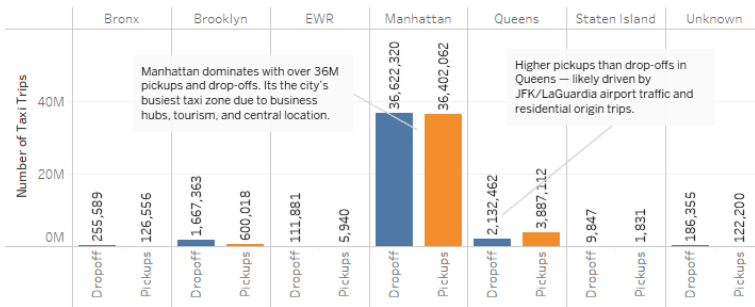
2024 NYC Taxi Demand Patterns: Daily Trends, Holiday Dips & Active Driver Benchmark



Seasonal Mobility Patterns in NYC: Taxi vs Subway (2024)

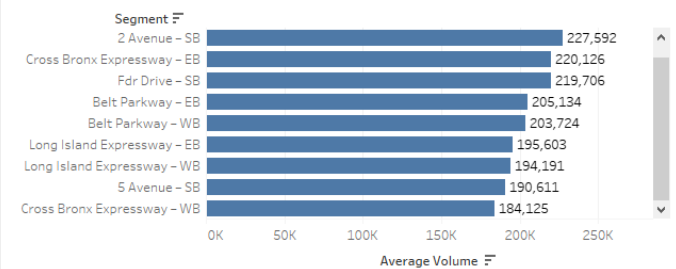


Borough-wise Taxi Pickup vs Dropoff Comparison (2024)



Top 10 Congested Road Segments in NYC (2024)

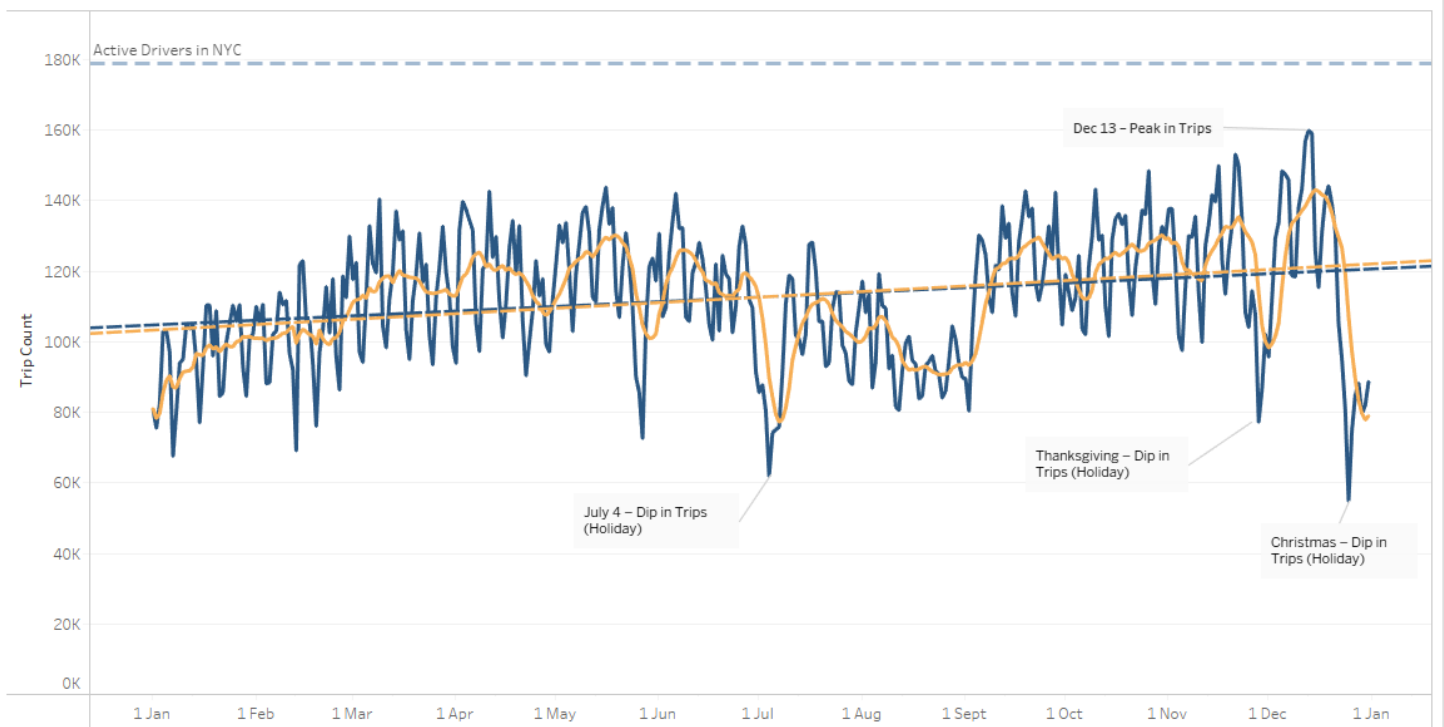
Based on average vehicle volume



This dashboard sets the foundation by offering a panoramic view of mobility trends across NYC throughout 2024. Each chart builds context for how the city moves across time, space, and infrastructure.

Chart: Daily Taxi Trips with Trends

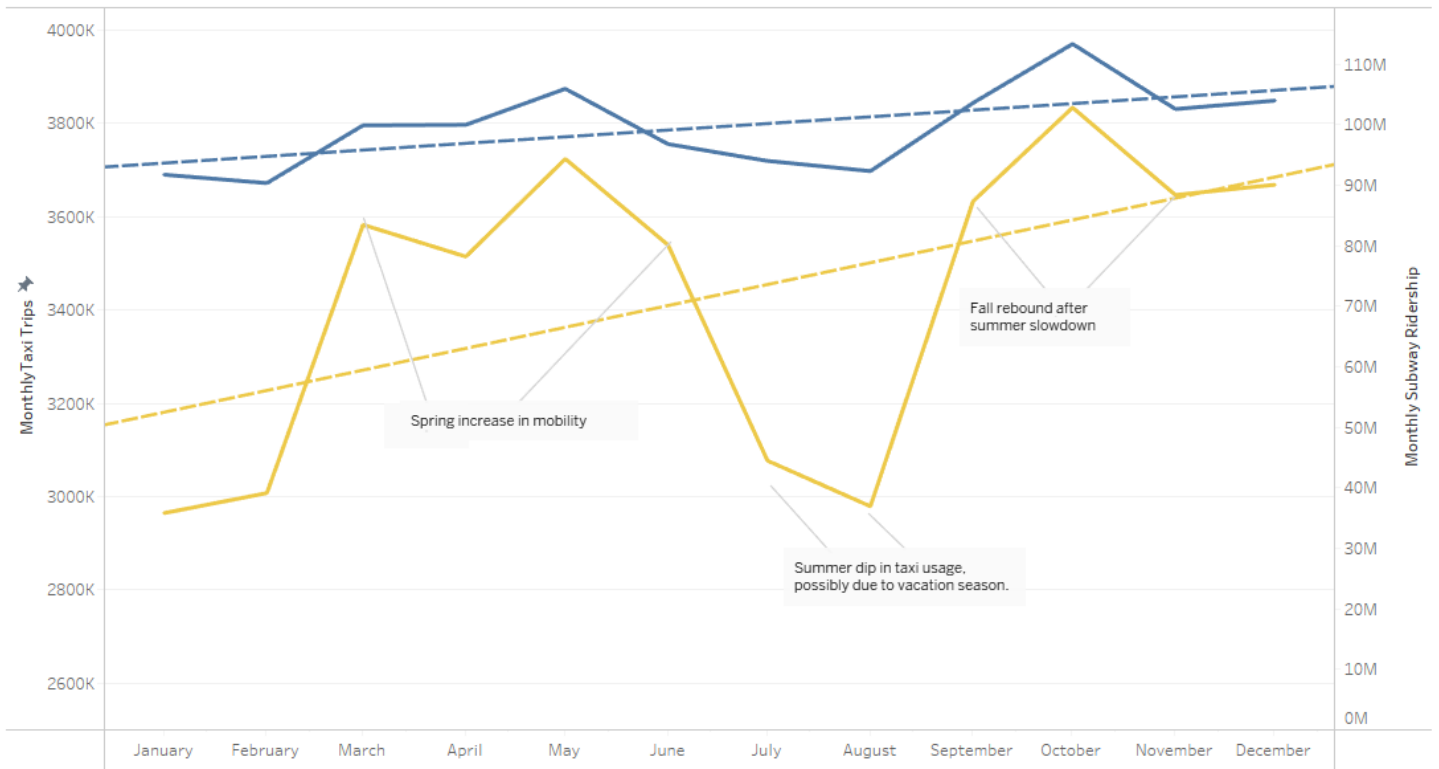
2024 NYC Taxi Demand Patterns: Daily Trends, Holiday Dips & Active Driver Benchmark



This chart highlights the city's consistent weekday travel rhythm, with trip counts regularly exceeding **50,000 per day** during busy weeks. Holiday dips like July 4, Thanksgiving, and Christmas stand out as moments of sharp decline. The overall trend line points upward, suggesting that taxi ridership grew steadily over the year. A **moving average line** helps smooth out anomalies and draws attention to gradual changes in demand. We also introduced a reference line showing **178,834** active medallion drivers, helping us contextualize whether supply was meeting growing demand.

Chart: Monthly Taxi vs Subway Ridership

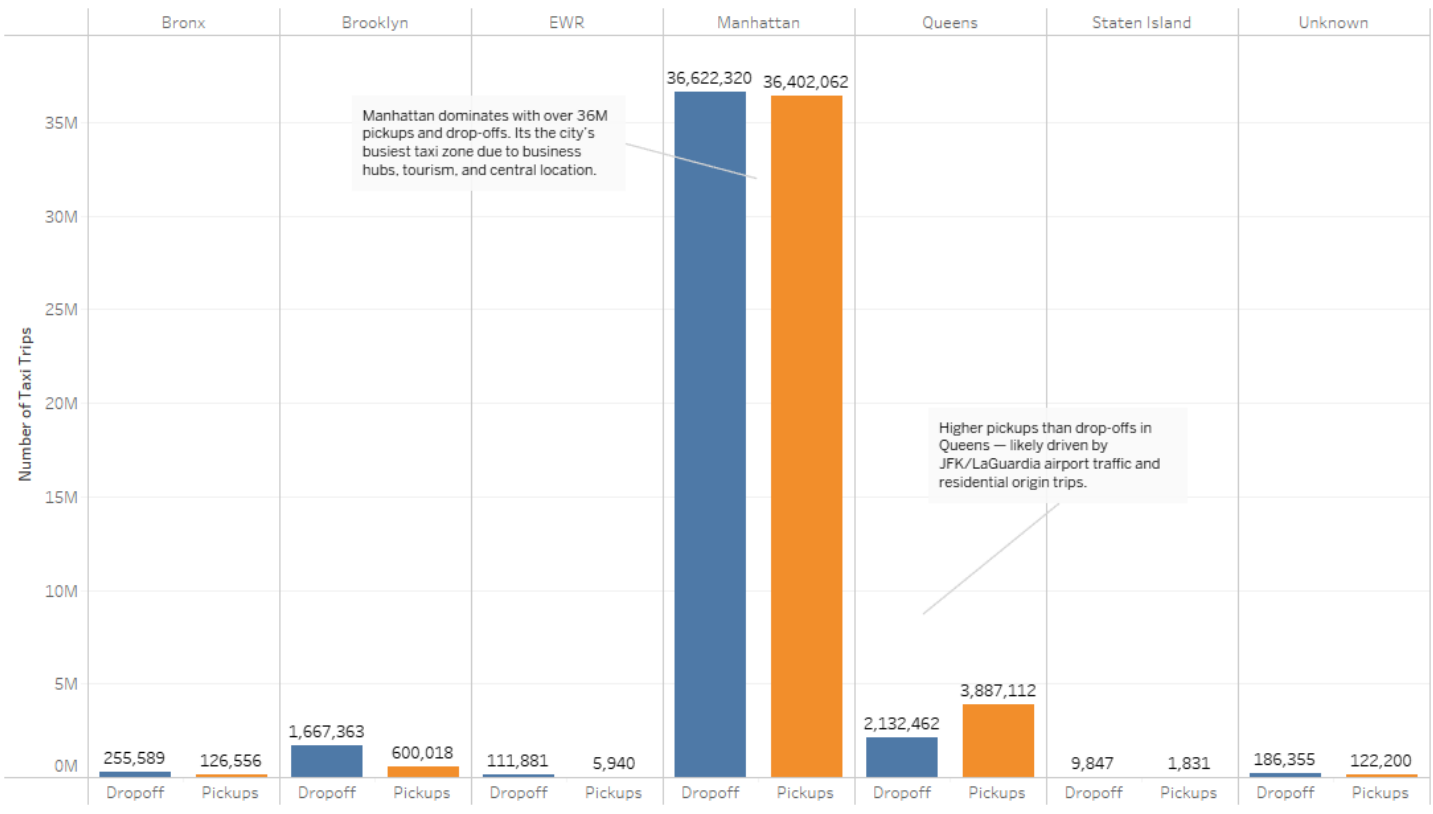
Seasonal Mobility Patterns in NYC: Taxi vs Subway (2024)



This chart compares modal preferences between taxis and the subway. Both systems show high usage in **May** and **October**, while **August** sees a clear dip in both indicating summer slowdown. Notably, **December** shows a strong spike in taxi ridership, narrowing the gap with subway volume and hinting at higher reliance on taxis during colder or more festive months.

Chart: Dropoffs vs Pickups by Borough

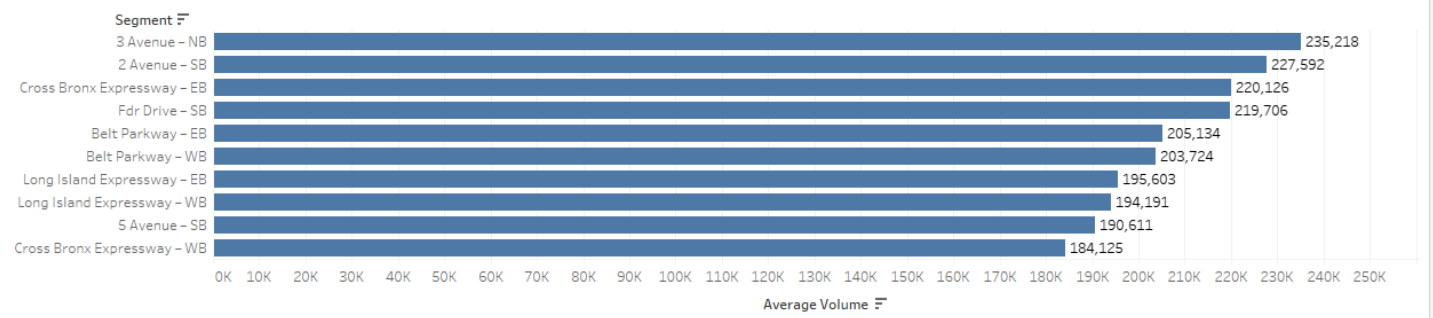
Borough-wise Taxi Pickup vs Dropoff Comparison (2024)



A spatial view of demand, this chart shows that **Manhattan** accounts for **over 75%** of both pickups and drop-offs. **Queens** has more pickups than drop-offs, likely reflecting outbound airport trips from JFK or LaGuardia. **Brooklyn and the Bronx** show lower volumes but are vital in framing the reach of NYC’s taxi system beyond central business areas.

Chart: Congested Road Segments NYC

Top 10 Congested Road Segments in NYC (2024)
Based on average vehicle volume

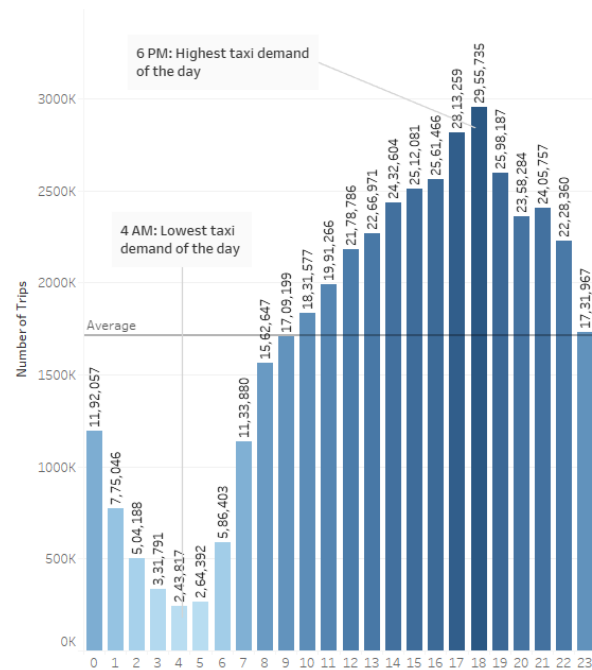


This chart surfaces NYC’s most strained road segments. The **Cross Bronx Expressway** and **Long Island Expressway** routinely exceed **350,000 vehicles**. These arteries are crucial to city flow but also hotspots for traffic bottlenecks. Overlaying this with taxi volume helps contextualize driver delays and route inefficiencies.

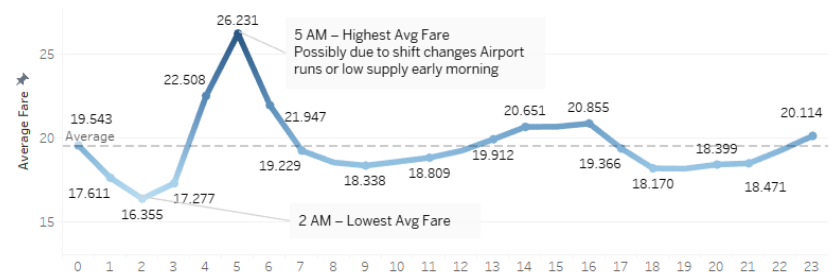
Dashboard 2: Hourly Dynamics – Demand vs Pricing (2024)

Hourly Demand vs Fare: Are Riders Paying a Fair Price?

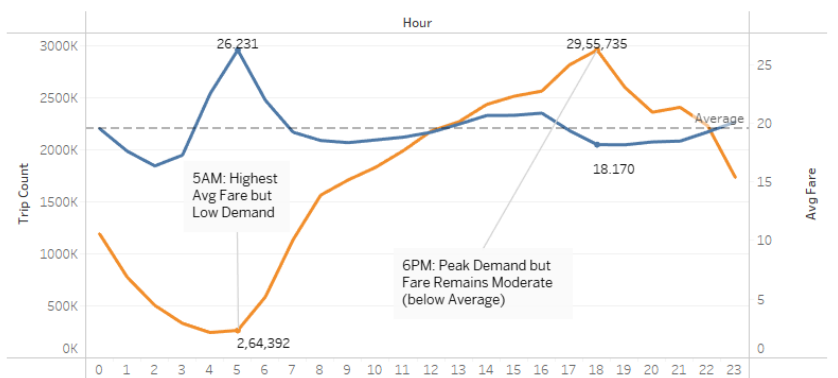
Hourly Taxi Usage in NYC (Hour of Day: 0–23) – Rush Hour Patterns (2024)



When Is It Most Expensive to Ride a Taxi in NYC? (2024)



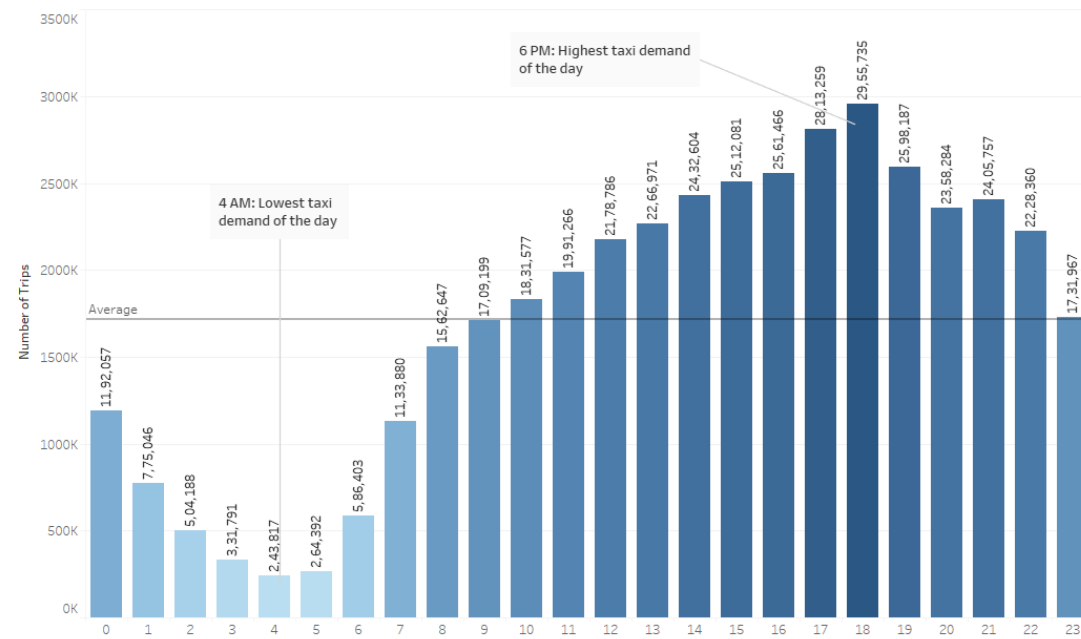
Hourly Taxi Fare vs Trip Demand in NYC (2024) – Are Prices Aligned With Demand?



This dashboard provides a deep dive into **how NYC moves throughout the day** and how fare pricing responds to that movement.

Chart: Hourly Taxi Trip Demand

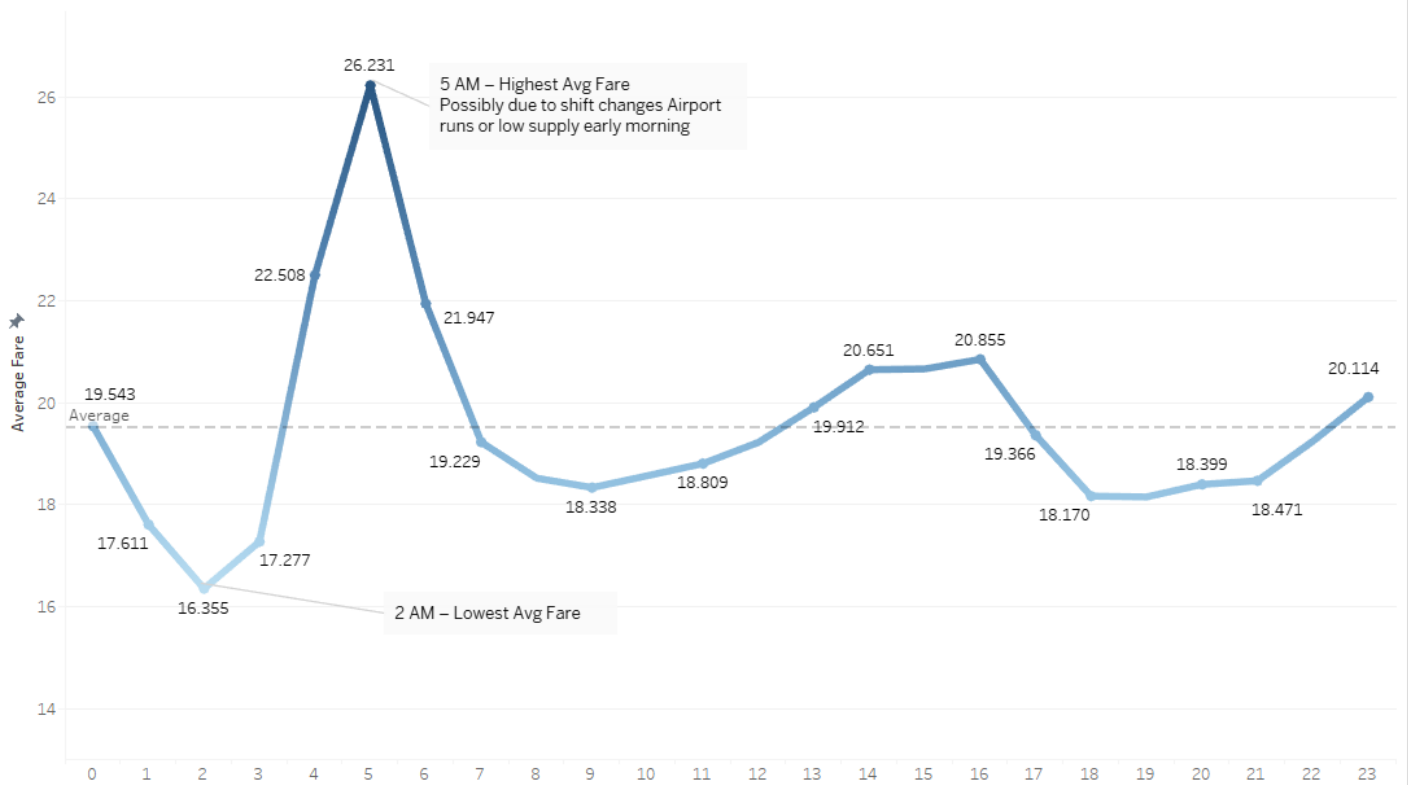
Hourly Taxi Usage in NYC (Hour of Day: 0–23) – Rush Hour Patterns (2024)



Trip volume is lowest between **2 AM and 5 AM**, then climbs steadily to a **6 PM peak** exceeding **29,00,000 trips**. This pattern aligns with commuter behavior and evening leisure activity. The relatively flat late-night hours reflect reduced rider activity.

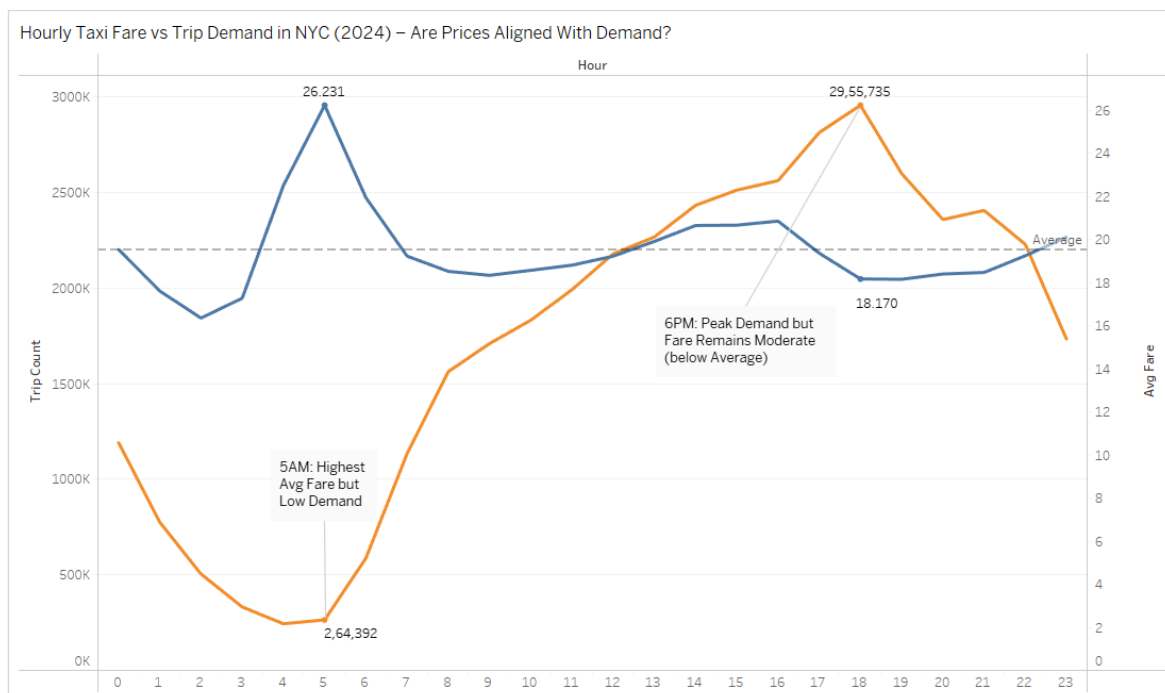
Chart: Hourly Taxi Avg Fare (2024)

When Is It Most Expensive to Ride a Taxi in NYC? (2024)



Fares reach their highest point at **5 AM** averaging **\$26** despite demand being relatively low. This unusual pricing suggests that **scarcity of drivers**, longer airport trips, or lack of public transit options at that hour may be inflating prices. This phenomenon continues slowly declines until **7 AM**, after which prices stabilize.

Chart: Hourly Fare vs Trip Volume



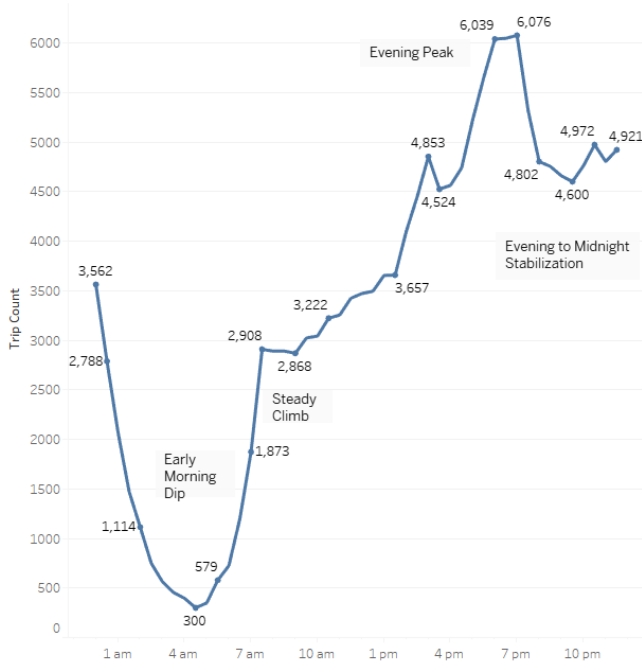
Overlaying fare and volume show a **mismatch**: while demand peaks during evening rush (6 PM), fare levels are moderate. Conversely, high fares during early morning hours do not match trip counts. These mismatches may point to inefficiencies in pricing algorithms or underutilized surge windows.

Dashboard 3: Dec 13 Taxi Behaviour

Understanding Peak-Day Mobility through Trips, Fares, and Distance

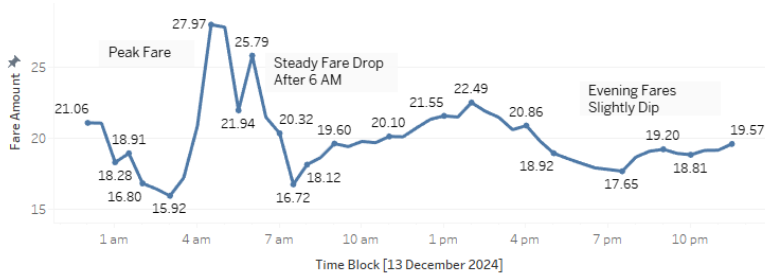
Trip Volume on December 13, 2024

Visualizing half-hourly trip counts throughout the peak day



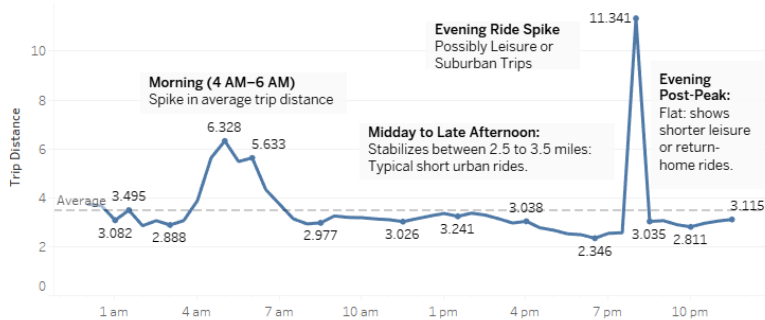
Fare Trends on December 13, 2024

How Much Do Riders Pay Throughout the Day?



Trip Distance Trends on December 13, 2024

How Far Did People Travel Throughout the Day?



Zooming in on **December 13** — the busiest day of the year — this dashboard examines the taxi system's behavior across that single high-demand day.

Chart: Trip Volume - Dec 13

Trip Volume on December 13, 2024

Visualizing half-hourly trip counts throughout the peak day



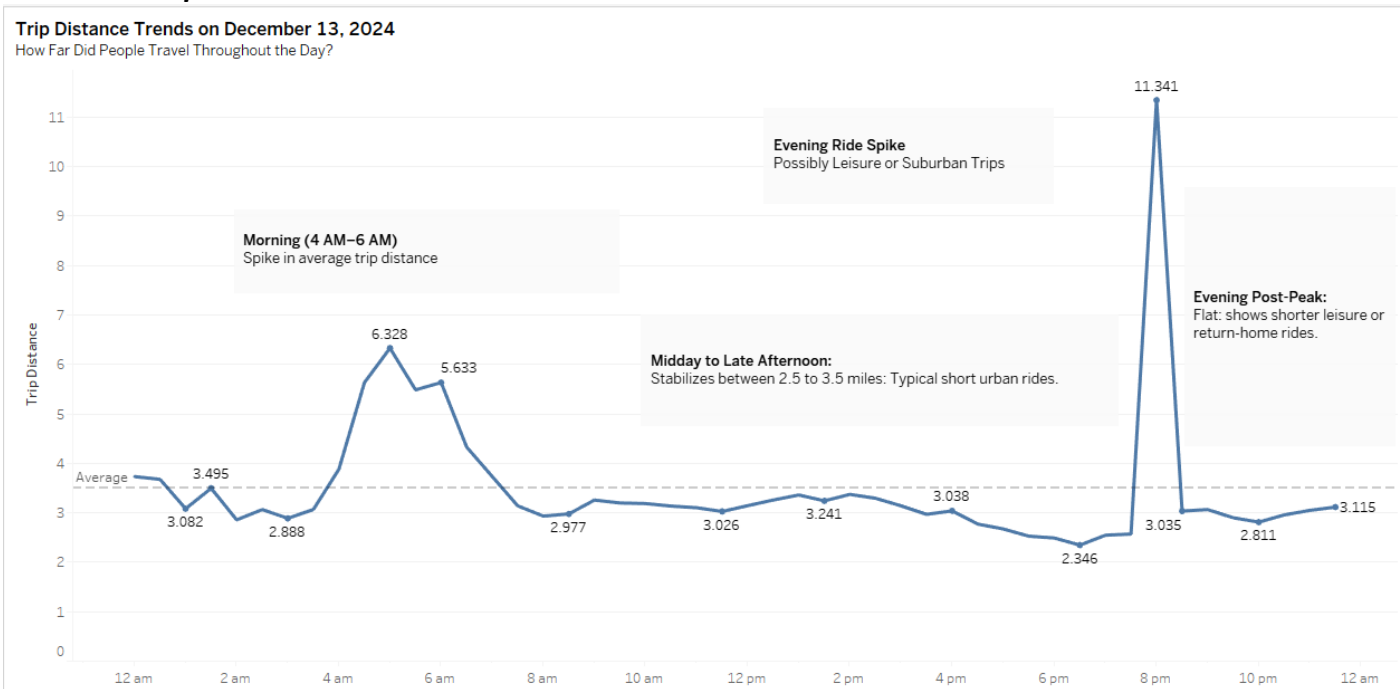
Starting from early morning, volume grows gradually with a clear spike at **6 PM** crossing **6,000 trips**, consistent with the overall annual pattern. The half-hour granularity shows the **ebb and flow** of mobility with surges around **lunch hours** and post-work commutes.

Chart: Fare Trends - Dec 13



The average fare peaks early between **4–6 AM**, above **\$27**, likely due to longer-distance trips or limited supply. From midday onward, pricing remains stable, even during high demand, which may indicate **ineffective fare scaling** or **driver saturation** absorbing demand.

Chart: Trip Distance - Dec 13



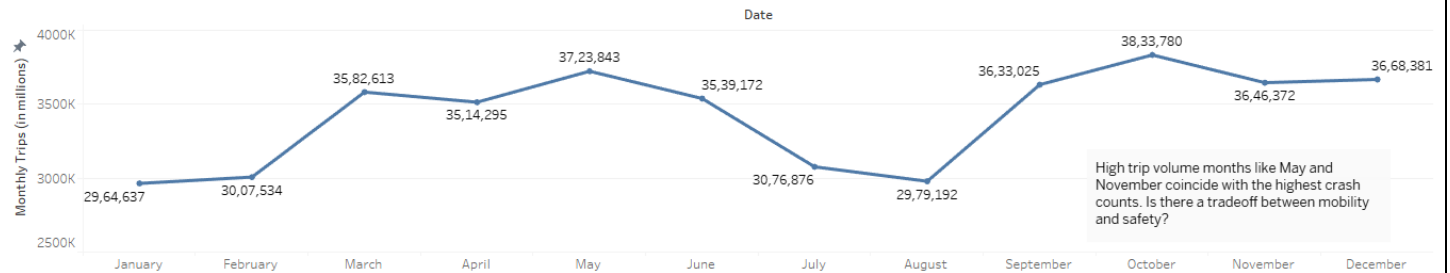
Trips on this peak day mostly fall within **2–3.5 miles**, but early morning and late evening show spikes in distance to around **6 miles**, again hinting at airport transfers or longer cross-borough travel.

Dashboard 4: Mobility vs Safety

Mobility vs Safety – Are More Trips Riskier?

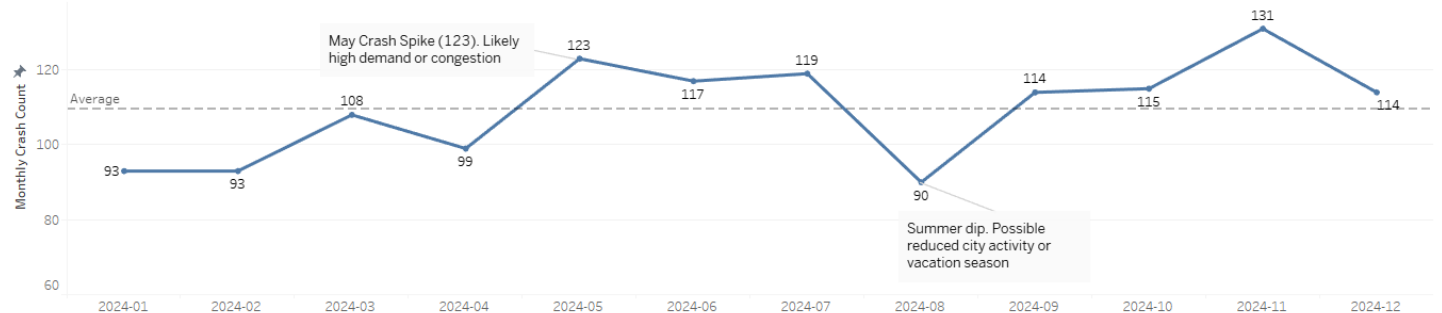
Exploring how monthly taxi activity may be linked to accident frequency

Monthly Taxi Trip Distribution: NYC (2024)



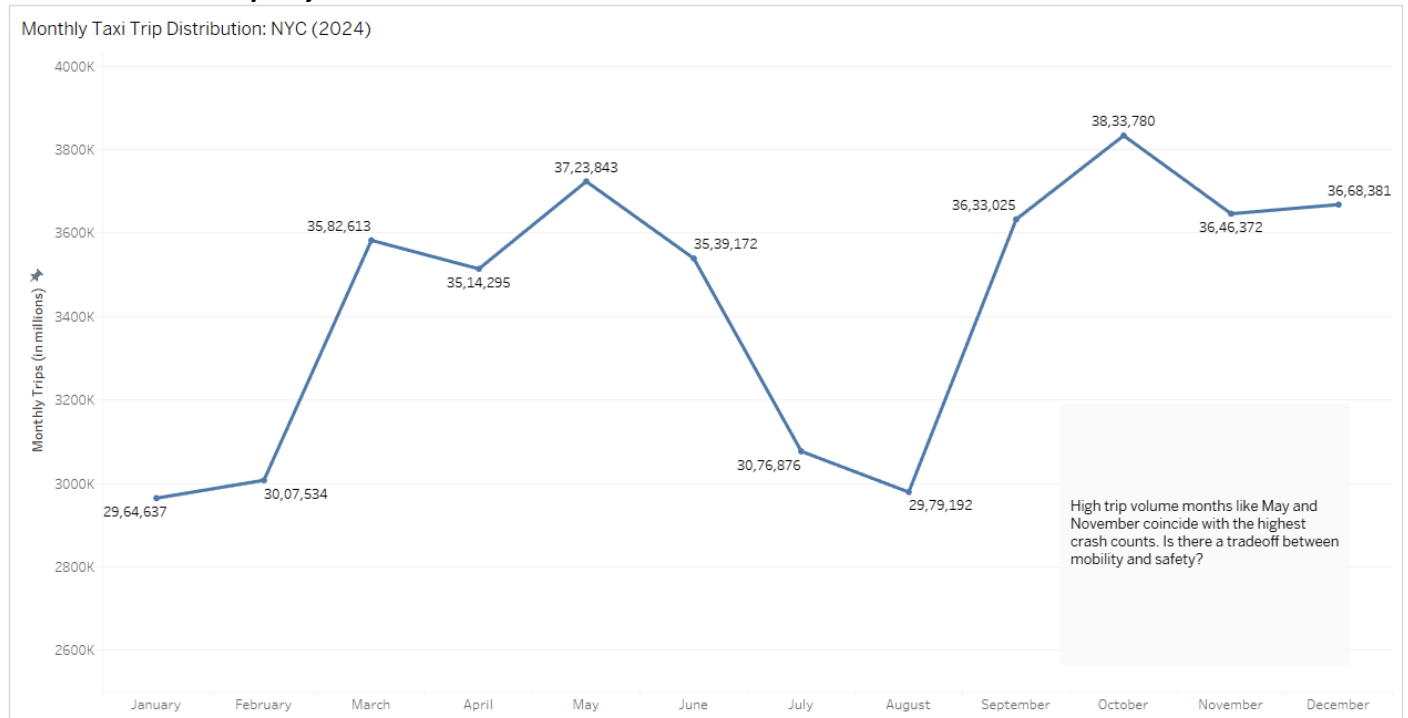
Monthly Crash Involvement of Yellow Taxis in NYC (2024)

Spike in May and November — Possible correlation with high mobility months



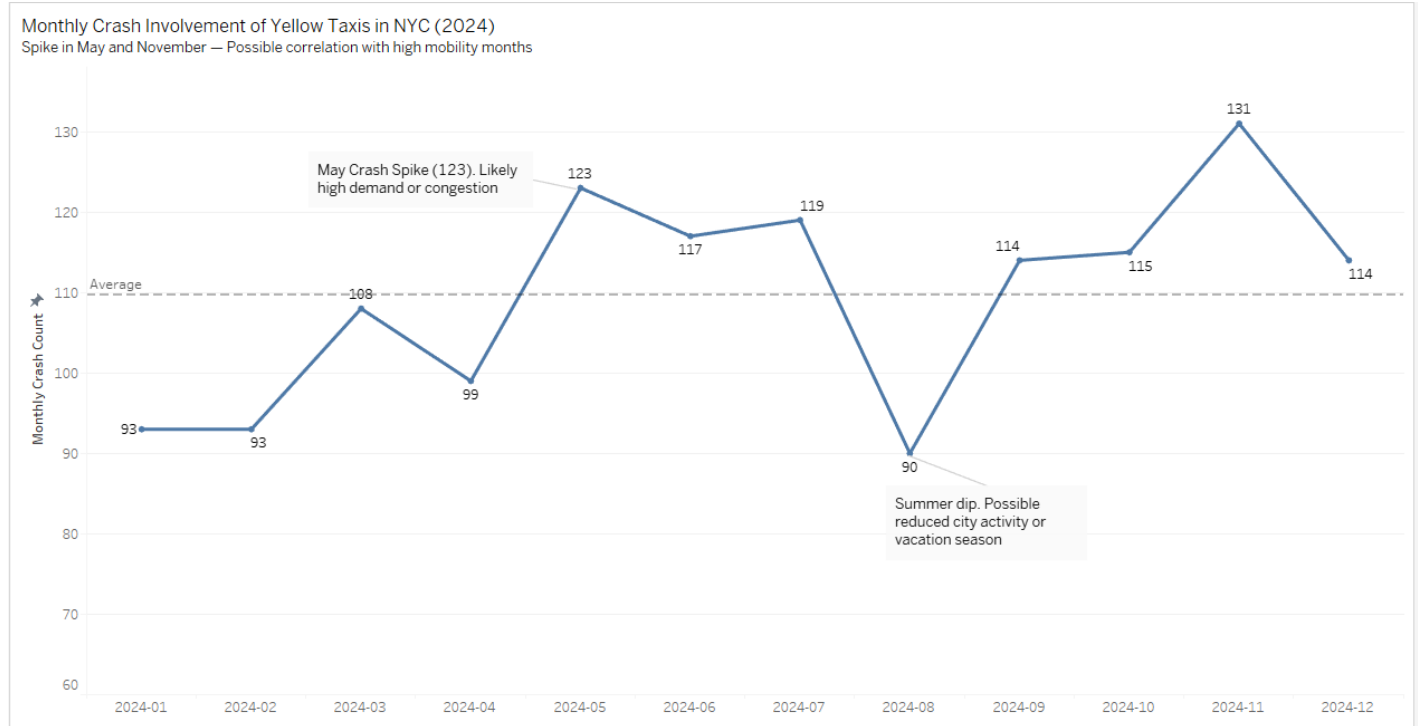
This dashboard explores the relationship between how much NYC moves and the number of taxi-involved crashes.

Chart: Taxi Trips by Month



Trip volume builds toward a **May peak (3.7 million)**, dips during **August**, and rises again in **November and December**. These cycles mirror commuter patterns, school breaks, and holiday travel.

Chart: Crash Trend Over Time 2024



The number of taxi crashes **peaks in May (123 incidents)** and again in **November (131)**, aligning with high-mobility months. In contrast, **August**, which had the lowest mobility, also saw the fewest crashes (**90**). These patterns suggest a **clear correlation between ride volume and incident frequency**, raising questions about citywide safety measures during high-demand periods.

Conclusion

Our analysis reveals how NYC's taxi system is shaped by a dynamic mix of rider demand, driver availability, pricing structures, and infrastructure pressure points. In 2024, despite **over 178,000 active drivers**, peak hours still showed fare-demand mismatches. The **highest single-day trip volume on Dec 13** underlined the system's elasticity, while patterns of under-pricing during high demand could suggest areas for policy intervention.

Crash trends mirrored ride volume, especially in **May and November**, reinforcing that more traffic often comes with more risk. From traffic bottlenecks in the Cross Bronx to long-distance morning trips to the airport, every data point contributes to a clearer picture of how NYC moves and where it can improve.

This project underscores the value of using visual analytics to guide decisions in pricing policy, public safety, and infrastructure investment. Future extensions could include predictive modeling of crashes, zone-level fare optimization, or the integration of ride-share data for a more holistic mobility lens.